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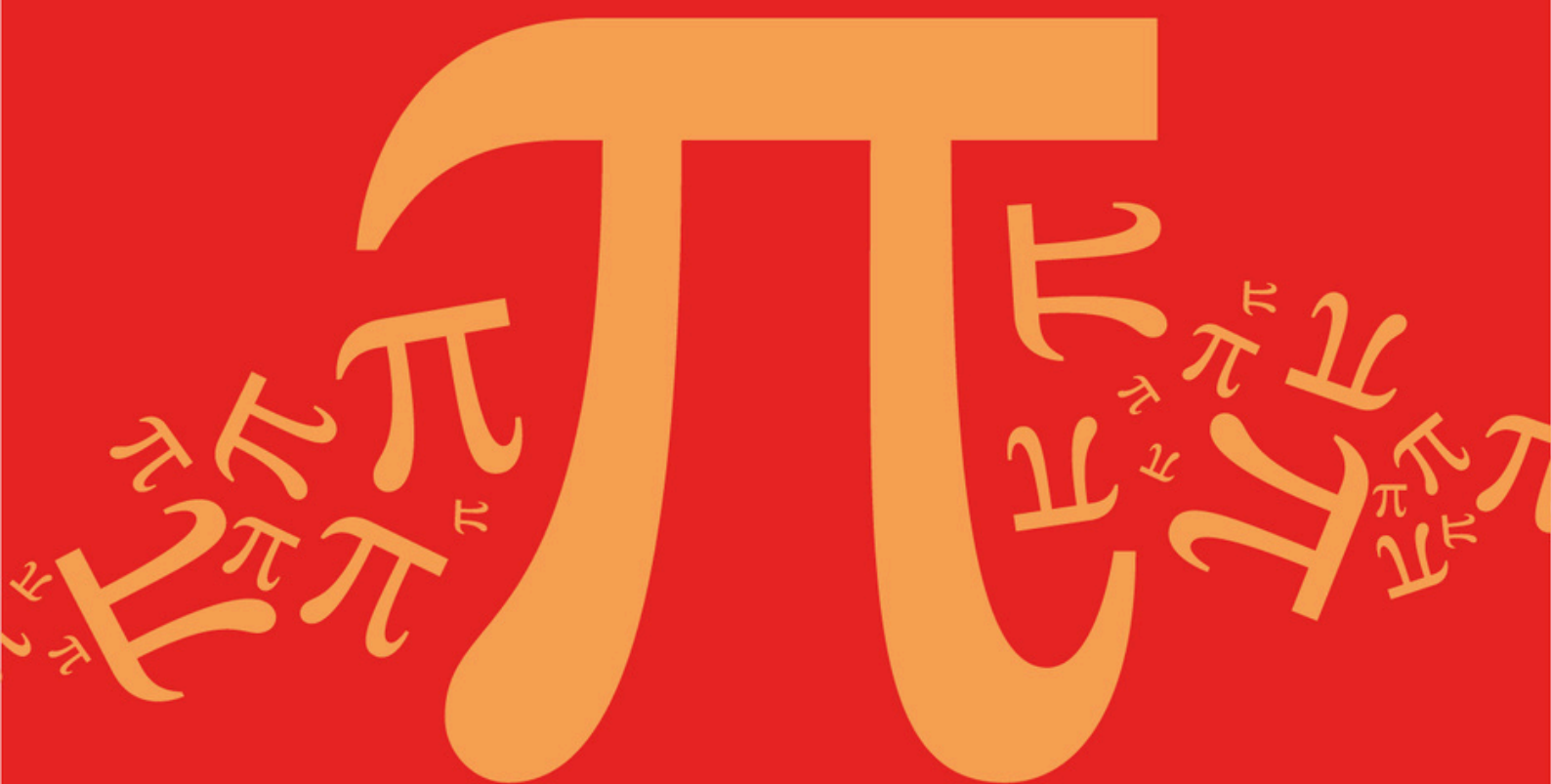
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Detailed mark scheme

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AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D2: Transform Polynomial Equation

Mark Scheme

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D2: Transform Polynomial Equation

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

(a) $\alpha + \beta = -2$

B1

$\alpha\beta = -5$

B1

2

(b) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-2)^2 - 2(-5)$

OE Using correct identity for $\alpha^2 + \beta^2$ with ft or correct substitution

M1

$= 14$

CSO A0 if $\alpha + \beta$ has wrong sign

A1

2

(c) $\alpha^3\beta + \alpha\beta^3 = \alpha\beta(\alpha^2 + \beta^2)$

PI Seen at least once in part (c).

OE eg $\alpha^3\beta + \alpha\beta^3 = \alpha\beta[(\alpha + \beta)^2 - 2\alpha\beta]$

M1

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$S(\text{um}) = \alpha^3\beta + \alpha\beta^3 + 2 = (-5)(14) + 2 = -68$

Correct or ft c's $\alpha\beta \times$ c's [answer (b)] + 2

A1F

$P(\text{roduct}) = (\alpha\beta)^4 + \alpha^3\beta + \alpha\beta^3 + 1 = (-5)^4 + (-5)(14) + 1 = 556$

Correct or

ft [c's $\alpha\beta$]⁴ + c's $\alpha\beta \times$ c's [answer (b)] + 1

A1F

$x - Sx + P (=0)$

Using correct general form of LHS of eqn **with** ft substitution of c's S and P values.

M1

Eqn.: $x^2 + 68x + 556 = 0$

CSO ACF

A1

5

Q2.

(a) $\alpha + \beta = -\frac{3}{2}$

OE

B1

$\alpha\beta = -3$

OE

B1

2

(b) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

Using correct identity for $\alpha^3 + \beta^3$ in terms of $\alpha + \beta$ and $\alpha\beta$.

M1

$= \left(-\frac{3}{2}\right)^3 - 3(-3)\left(-\frac{3}{2}\right)$

with ft / or correct substitution

A1F

$= -\frac{27}{8} - \frac{27}{2} = -\frac{135}{8}$

CSO AG. Correct evaluation of each of $(-1.5)^3$ and $-3(-3)(-1.5)$ must be seen before the printed answer is stated

A1

3

(c) Sum = $\alpha + \frac{\alpha}{\beta^2} + \beta + \frac{\beta}{\alpha^2} = \alpha + \beta + \frac{\alpha^3 + \beta^3}{(\alpha\beta)^2} = -\frac{3}{2} + \frac{-135/8}{9}$

Writing $\alpha + \frac{\alpha}{\beta^2} + \beta + \frac{\beta}{\alpha^2}$ in a suitable form with ft/or correct substitution

M1

Sum = $-\frac{27}{8}$

PI OE exact value eg -3.375 (A0 if $\alpha\beta = 3$ used to get $(\alpha\beta)^2 = 9$)

A1

Product = $\alpha\beta + \frac{\beta}{\alpha} + \frac{\alpha}{\beta} + \frac{1}{\alpha\beta} = \alpha\beta + \frac{\alpha^2 + \beta^2}{\alpha\beta} + \frac{1}{\alpha\beta}$ (*)

Now $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta (= \frac{9}{4} + 6)$

(*) OE with correct identity for $\alpha^2 + \beta^2$ used in (c). Subst of values not required but PI by correct value of Product

M1

Product = $-3 - \frac{1}{3}\left(\frac{9}{4} + 6\right) - \frac{1}{3} = -\frac{73}{12}$
PI OE exact value

A1

$x^2 - Sx + P (= 0)$

Using correct general form of LHS of eqn **with** ft substitution of c's S and P values.

M1

Eqn is $24x^2 + 81x - 146 = 0$

OE but integer coefficients and ' $= 0$ ' needed

A1

6

[11]

Q3.

(a) $\alpha + \beta = -\frac{7}{2}$

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B1

$\alpha\beta = 4$

B1

(b) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{7}{2}\right)^2 - 2(4)$

Using correct identity with ft or correct substitution

M1

$= \frac{49}{4} - 8 = \frac{17}{4}$

CSO AG. A0 if $\alpha + \beta$ has wrong sign

A1

2

(c) (Sum =)

$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{17/4}{16} \left(= \frac{17}{64} \right)$

Writing $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ in a correct suitable form with ft or correct substitution

M1

$$= \frac{17}{64}$$

ft wrong value for $\alpha\beta$

A1F

(Product =) $\frac{1}{(\alpha\beta)^2} = \frac{1}{16} \left(= \frac{4}{64} \right)$
 ft wrong value for $\alpha\beta$

B1F

$$x^2 - Sx + P (= 0)$$

Using correct general form of LHS of eqn **with ft** substitution of c's S and P values. PI

M1

Eqn is $64x^2 - 17x + 4 = 0$

CSO Integer coefficients and '= 0' needed

A1

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[9]

Q4.

Accept correct equivalent decimals in place of some / all fractions in the scheme

(a) $\alpha + \beta = \frac{7}{5} (= 1.4)$

B1

$$\alpha\beta = \frac{1}{5} (= 0.2)$$

B1

2

(b) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

OE eg $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\frac{1}{5}[7(\alpha + \beta) - 1 - 1]}{\alpha\beta}$ scores M1 m1

M1

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{\left(\frac{7}{5}\right)^2 - 2\left(\frac{1}{5}\right)}{\frac{1}{5}}$$

Correct expression for $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ in terms of either $(\alpha + \beta)$ and $\alpha\beta$ or with numerical substitution of correct / c's values from (a)

m1

$$= \frac{\frac{49}{25} - 2\left(\frac{1}{5}\right)}{\frac{1}{5}} = \frac{\frac{49}{25} - \frac{2}{5}}{\frac{1}{5}} = \frac{\frac{39}{25}}{\frac{1}{5}} = \frac{39}{5}$$

CSO AG must see some intermediate evaluation, must see one of the first three expressions

A0 if $\alpha + \beta$ has wrong sign

A1

3

(c) (Sum =) $\alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta} = \alpha + \beta + \frac{\alpha + \beta}{\alpha\beta}$

Writing $\alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta}$ in a correct suitable form or with numerical values

M1

$$\left(= \frac{7}{5} + \frac{1}{\frac{1}{5}} \right)$$

(Product =) $\alpha\beta + \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + \frac{1}{\alpha\beta} = \frac{1}{5} + \frac{39}{5} + 5$

Correct expression for product into which substitution of numbers attempted for all terms, at least one either correct / correct ft

OE Both

M1

Sum = $\frac{42}{5}$, Product = 13

SC If B0 for $\alpha + \beta = -\frac{7}{5}$ in (a), and (c)

$S = -\frac{42}{5}$ oe, $P = 13$ award this A1

A1

$$x^2 - Sx + p (= 0)$$

Using correct general form of LHS of equation **with** ft substitution of c's S and P values. **PI**. **M0** if either $S = \alpha + \beta$ or $P = \alpha\beta$ values

M1

Equation is $5x^2 - 42x + 65 = 0$

CSO Integer coefficients and ' $= 0$ ' needed. Dependent on B1B1 in (a) and previous 4 marks in (c) scored

A1

5

[10]

Q5.

(a) (i) $\alpha + \beta + \gamma = 0$

B1

1

(ii) $\alpha\beta\gamma = -q$

B1

1

(b) $\alpha^3 + p\alpha + q = 0$

M1

$$\sum \alpha^3 + p \sum \alpha + 3q = 0$$

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m1

$$\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$

AG

A1

Alternative:

Use of $(\sum \alpha)^3 = (\sum \alpha^3) + 6\alpha\beta\gamma + 3(\sum \alpha \sum \alpha\beta - 3\alpha\beta\gamma)$

(M1)

Substitution of $\sum \alpha = 0$

(m1)

Result

(A1)

3

(c) (i) $\beta = 4 - 7i, \gamma = -8$

B1,B1

2

(ii) Attempt at either p or q

M1

$$p = 1$$

A1F

$$q = 520$$

ft incorrect roots provided p and q are real

A1F

3

(d) Replace z by $\frac{1}{z}$ in cubic equation

$$\text{or } \sum \frac{1}{\alpha} = -\frac{p}{q}, \sum \frac{1}{\alpha\beta} = 0, \frac{1}{\alpha\beta\gamma} = -\frac{1}{q}$$

ft on incorrect p and / or q

M1
A1F

$$520z^3 + z^2 + 1 = 0 \text{ coefficients must be integers}$$

CAO

A1

3

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[13]

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Q6.

(a) $\alpha + \beta = 6, \alpha\beta = 18$

B1B1

2

(b) Sum of new roots = $6^2 - 2(18) = 0$
ft wrong value(s) in (a)

M1A1F

$$\text{Product} = 18^2 = 324$$

ditto

B1F

$$\text{Equation } x^2 + 324 = 0$$

'= 0' needed here;

ft wrong value(s) for sum/product

A1F

4

- (c) α^2 and β^2 are $\pm 18i$

B1

1

[7]

Q7.

(a) $\alpha + \beta = -\frac{3}{2}, \alpha\beta = \frac{3}{4}$

B1B1

2

(b) $\alpha^2 + \beta^2 = \left(-\frac{3}{2}\right)^2 - 2\left(\frac{3}{4}\right) = \frac{3}{4}$

AG; A0 if $\alpha + \beta$ has wrong sign

M1A1

2

(c) Sum = $2(\alpha + \beta) = -3$

ft wrong value for $\alpha + \beta$

B1F

Product = $10\alpha\beta - 3(\alpha^2 + \beta^2) = \frac{21}{4}$

ft wrong values

M1A1F

$x^2 - Sx + P (= 0)$

Signs must be correct for the M1

M1

Eqn is $4x^2 + 12x + 21 = 0$

Integer coeffs and ' $= 0$ ' needed

A1

5

[9]

Q8.

(a) $\alpha + \beta = 2, \alpha\beta = \frac{1}{3}$

B1B1

2

(b) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
or other appropriate formula

M1

$$\dots = 8 - 3\left(\frac{1}{3}\right)(2) = 6$$

*m1 for substn of numerical values;
 A1 for result shown (AG)*

m1A1

3

(c) Sum of roots = $\frac{\alpha^3 + \beta^3}{\alpha\beta}$

M1

$$\dots = \frac{6}{\frac{1}{3}} = 18$$

ft wrong value for $\alpha\beta$

A1F

Product = $\alpha\beta = \frac{1}{3}$
ditto

B1F

Equation is $3x^2 - 54x + 1 = 0$

*Integer coeffs and “= 0” needed;
 ft wrong sum and/or product*

A1F

4

[9]

Q9.

(a) $\alpha + \beta = 4, \alpha\beta = 10$

B1,B1

2

(b) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$

M1

$$= \frac{4}{10} = \frac{2}{5}$$

convincingly shown (AG)

A1

2

(c) Sum of roots = $(\alpha + \beta) + 2(\text{ans to (b)})$

M1

$$= 4\frac{4}{5}$$

ft wrong value for $\alpha + \beta$

A1F

Product = $\alpha\beta + 4 + \frac{4}{\alpha\beta}$

*M1 for attempt to expand product
(at least two terms correct)*

M1A1

$$= 14\frac{2}{5}$$

ft wrong value for $\alpha\beta$

A1F

Equation is $5x^2 - 24x + 72 = 0$

*integer coeffs and ' $= 0$ ' needed here;
ft one numerical error*

A1F

6

[10]

Q10.

(a) $\alpha + \beta = -\frac{1}{2}, \alpha\beta = -4$

B1B1

2

(b) $\alpha^2 + \beta^2 = \left(-\frac{1}{2}\right)^2 - 2(-4) = 8\frac{1}{4}$

*M1 for substituting in correct formula;
ft wrong answer(s) in (a)*

M1A1F

2

(c) Sum of roots = $4\left(8\frac{1}{4}\right) = 33$

ft wrong answer in (b)

B1F

Product = $16(\alpha\beta)^2 = 256$
ft wrong answer in (a)

B1F

Equation is $x^2 - 33x + 256 = 0$
ft wrong sum and/or product;
allow ' $p = -33, q = 256$ ';
condone omission of ' $= 0$ '

B1F

3

[7]

Q11.

(a) (i) $\alpha + \beta = 2, \alpha\beta = 4$

B1B1

$$\alpha^3 + \beta^3 = (2)^3 - 3(4)(2) = -16$$

M1A1

$\alpha^3 \beta^3 = (4)^3 = 64$, hence result
convincingly shown (AG)

M1A1

6

(ii) Discriminant 0, so roots equal
or by factorisation

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B1E1

2

(b) $x = \frac{2 \pm \sqrt{4 - 16}}{2}$

or by completing square

M1

$$\dots = 1 \pm \frac{1}{2}i\sqrt{12}$$

A1

2

(c) $\alpha, \beta = 1 \pm i\sqrt{3}$
 and $\alpha^3 = \beta^3$, hence result

E2

2

[12]

Q12.

(a) (i) $\sum \alpha = -i$

B1

1

(ii) $\sum \alpha\beta = 3$

B1

1

(iii) $\alpha\beta\gamma = 1 + i$

B1

1

(b) (i) $\sum \alpha^2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta$ *used*
Allow if sign error or 2 missing

M1

$= (-i)^2 - 2 \times 3$

A1F

$= -7$

ft errors in (a)

A1F

3

(ii) $\sum \alpha^2 \beta^2 = \left(\sum \alpha\beta\right)^2 - 2\sum \alpha\beta \cdot \beta\gamma$
Allow if sign error in 2 missing

M1

$= \left(\sum \alpha\beta\right)^2 - 2\alpha\beta\gamma \sum \alpha$

A1

$= 9 - 2(1 + i)(-i)$

ft errors in (a)

A1F

$= 7 + 2i$

ft errors in (a)

A1F

4

(iii) $\alpha^2 \beta^2 \gamma^2 = (1 + i)^2 = 2i$

M1

ft sign error in $\alpha\beta\gamma$

A1F

2

(c) $z^3 + 7z^2 + (7 + 2i)z - 2i = 0$

Correct numbers in correct places

B1F

Correct signs

B1F

2

[14]

Q13.

(a) $\alpha + \beta = -1, \alpha\beta = 5$

B1B1

2

(b) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

with numbers substituted

M1

$\dots = 1 - 10 = -9$

ft sign error(s) in (a)

A1F

2

(c) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

M1

$\dots = -\frac{9}{5}$

AG: A0 if $\alpha + \beta = 1$ used

A1

2

(d) Product of new roots is 1

PI by constant term 1 or 5

B1

Eqn is $5x^2 + 9x + 5 = 0$

ft wrong value for product

B1F

2

[8]

Q14.

(a) $\alpha + \beta = \frac{1}{2}, \alpha\beta = 2$

B1B1

2

(b) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$

M1

$$\dots = \frac{\frac{1}{2}}{2} = \frac{1}{4}$$

Convincingly shown (AG)

A1

2

(c) Sum of roots = 1

PI by term $\pm x$; ft error(s) in (a)

B1F

Product of roots = $\frac{16}{\alpha\beta} = 8$

ft wrong value of $\alpha\beta$

B1F

Equation is $x^2 - x + 8 = 0$

ft wrong sum/product; “= 0” needed

B1F

3

[7]

Q15.

(a) $\alpha + \beta = 2, \alpha\beta = \frac{2}{3}$

SC 1/2 for answers 6 and 2

B1B1

2

(b) (i) $(\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$

Accept unsimplified

B1

1

(ii) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

M1

Substitution of numerical values

m1

$$\alpha^3 + \beta^3 = 4$$

convincingly shown AG

A1

3

(c) $\alpha^3\beta^3 = \frac{8}{27}$

B1

Equation of form $px^2 \pm 4px + r = 0$

M1

Answer $27x^2 - 108x + 8 = 0$

ft wrong value for $\alpha^3\beta^3$

A1ft

3

[9]

Q16.

(a) $\alpha + \beta = 5, \alpha\beta = -2$

B1, B1

2

(b) $\alpha^2\beta + \alpha\beta^2 = (\alpha + \beta) = -10$
ft wrong values

M1A1ft

2

(c) $(\alpha^2\beta)(\alpha\beta^2) = (\alpha\beta)^3 = -8$
ft wrong values

M1A1ft

Equation is $x^2 + 10x - 8 = 0$

Dep on both M1s; ft wrong values;
Condone omission of “= 0”

A1ft

3

[7]

Q17.

(a) (i) $\alpha + \beta = 4, \alpha\beta = 13$

B1B1

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\dots = 4^2 - 26 = -10$

convincingly shown (AG)

M1

A1

2

(iii) The square of a real number is positive (or zero)

E1

The sum of two such squares is positive (or zero)

E1

2

(b) (i) $(\alpha + i) + (\beta + i) = 4 + 2i$
ft wrong value in (a)(i)

B1F

1

(ii) $(\alpha + i)(\beta + i) = 12 + 4i$
ditto

M1A1F

2

(c) Correct coeff of x or constant term
Using c 's answers in (b)

M1

$x^2 - (4 + 2i)x + (12 + 4i) = 0$

ft wrong answers in (b)

A1F

2

[11]

Examiner reports

Q1.

This question, which tested roots and coefficients of a quadratic equation, proved to be a good source of marks for many students. Generally parts (a) and (b) were answered correctly. In part (c) a large majority of students used the appropriate identity $\alpha^3\beta + \alpha\beta^3 = \alpha\beta(\alpha^2 + \beta^2)$ to find the sum and products of the new roots correctly, although the error $(-5)^4 = -625$ was not a rarity in finding an incorrect product. Although not as common as in recent papers, the omission of ' $= 0$ ' from the final answer was seen.

Q2.

This question tested roots and coefficients of a quadratic equation. Almost all students wrote down the correct values in part (a). In part (b) a large proportion of students quoted and used the correct identity for $\alpha^3 + \beta^3$ but a significant number failed to score the final mark because they did not show sufficient evaluations before writing down the printed answer. With the answer given, examiners expected to see correct evaluations of each of $(-1.5)^3$ and $-3(-3)(-1.5)$ before the printed answer was stated. Average grade students generally picked up the method marks in part (c) but, in general, only the more able students could cope with the algebraic demands needed to obtain the correct quadratic equation. However, it was pleasing to see that there were fewer students than normal who missed off the ' $= 0$ ' from their quadratic equation.

Q3.

This opening question, which tested roots and coefficients of a quadratic equation, proved to be a good source of marks for almost all candidates.

With the answer given in part (b), examiners expected candidates to show some

evaluation between ' $(-3.5)^2 - 2(4)$ ' and the printed answer $\frac{17}{4}$.

In part (c), where a quadratic equation was asked for, most candidates applied a correct method but a number of solutions were seen which ended without an equation, the ' $= 0$ ' having been omitted.

Q4.

This opening question which tested roots and coefficients of a quadratic equation proved to be a good source of marks for almost all students. As indicated in the previous report for a similar question in this unit, with the answer given in part (b), examiners expected students to show more than just a full substitution followed by the printed answer. Many

students reached the correct result $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\left(\frac{7}{5}\right)^2 - 2\left(\frac{1}{5}\right)}{\frac{1}{5}}$ and so scored the first two marks but a significant minority then just equated this to the printed value. Examiners

expected to see at least $\left(\frac{7}{5}\right)^2$ evaluated correctly before the printed answer for the award of the final mark. In part (c), other than arithmetic errors, the most common loss of a mark was a failure to give an equation, the ' $= 0$ ' having been omitted.

Q5.

Part (a) was invariably correctly quoted. However, in part (b) responses fell into several categories. The most successful students used the fact that α , β , and γ were each roots of the given cubic equation and therefore each satisfied that equation. Others quoted the formula for $\sum \alpha^3$ and, provided this quotation was correct, usually went on to secure full marks. A third category was those students who attempted to obtain $\sum \alpha^3$ by considering $(\sum \alpha)^3$ but these rarely succeeded, finding the algebra involved too complicated to handle. Some solutions ended at this juncture due to students not realising that a cubic equation with real coefficients must have a pair of complex conjugate roots if one root is complex. Part (c), apart from the odd slip of sign, was well done. In part (d), those students who

replaced z by $\frac{1}{z}$ in the original cubic equation completed this part of the question quickly, but those who tried to find the sums and products of the reciprocals of α , β and γ usually ended up with at least one incorrect result.

Q6.

Most candidates started the paper in confident fashion, earning the first six marks with apparent ease, though some lost the sixth mark through failing to write ' $= 0$ ' to complete their equation in part (b). Relatively few candidates realised that they were being tested on their knowledge of complex numbers in part (c), and even those who did sometimes failed to obtain the one mark on offer by misidentifying the two correct roots of their equation.

Q7.

Almost all candidates earned the first four marks with no apparent difficulty. In part (c), there were small errors especially in finding the product of the roots of the required equation. In the forming of the equation itself, most candidates were aware of the need to reverse the sign of the sum of the roots to give the coefficient of x . Integer coefficients were usually obtained correctly, but, as mentioned above, many candidates failed to write the necessary ' $= 0$ ' at the end.

Q8.

Most candidates seemed to be very familiar with the techniques needed in this question. The formula for the sum of the cubes of the roots was either quoted confidently and correctly or worked out from first principles. Errors occurred mainly in part (c): algebraic errors in finding the sum or even the product of the roots of the required equation; errors in choosing which numerical values to substitute for $\alpha\beta$ or $\alpha + \beta$; and, very frequently, a failure to present the final answer in an acceptable form, with integer coefficients and the " $= 0$ " at the end.

Q9.

This was another well-answered question. The first two parts presented no difficulty to any reasonably competent candidate. In part (c) some candidates, faced with the task of finding the sum of the roots of the required equation, repeated the work done in part (b) rather than quoting the result obtained there. The expansion of the product of the new roots caused some unexpected difficulties, some candidates failing to deal properly with two terms which should have given them constant values. The final mark was often lost by a failure to observe the technical requirements spelt out in the question.

Q10.

Most candidates found this a very good introduction to the paper and gained full marks, or very nearly so. Errors in parts (a) and (b) were rare, and were nearly always sign errors rather than those caused by the use of an incorrect procedure.

In part (c) the most common mistake was to have a 4 instead of a 16 in the product of the roots of the required equation. Some candidates failed to see the connection with part (b) when finding the sum of the roots, but more often than not they still found the right value for this sum.

Q11.

This question proved to be an excellent source of marks for the majority of candidates, apart from the discriminating test provided by part (c). Many candidates seemed to be familiar with the techniques relating to the cubes of the roots of a quadratic equation, though some struggled to find the correct expression for the sum of the cubes of the roots, while others lost marks by not showing enough evidence in view of the fact that the required equation was printed on the question paper.

Parts (a)(ii) and (b) proved straightforward for most candidates, but only a few were able to give a clear explanation in part (c), where many candidates claimed that the original roots α and β must be equal.

Q12.

This question was probably the most popular question on this paper and certainly showed candidates well prepared to answer questions on this part of the specification. There were many fully correct solutions or correct apart from the odd sign error, the most common of which was to write down $(-i)^2$ as +1 instead of -1. If there was a major loss of marks, it

was usually in the inability of a candidate to evaluate $\sum \alpha^2 \beta^2$ and in this case the candidate started by considering $(\sum \alpha^2)^2$ only to find that the evaluation of $\sum \alpha^4$ posed a serious problem.

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Q13.

The great majority of candidates showed a confident grasp of the algebra needed to deal with the sum of the squares of the roots of a quadratic equation, and answered all parts of this question efficiently. The only widespread loss of credit came in the very last part, where many candidates failed to give integer coefficients or else found a quadratic expression but without the necessary '=' to make it an equation.

Q14.

Most candidates answered this question very well. Only a small number of candidates gave the wrong sign for the sum of the roots at the beginning of the question. Rather more made the equivalent mistake at the end of the question, giving the x term with the wrong sign. Another common mistake at the end of the question was to omit the 'equals zero', so that the equation asked for was not given as an equation at all.

Q15.

This opening question provided most candidates with a very good start to the paper, with full marks or nearly full marks earned. It was pleasing to see that almost all candidates gave the correct values in part (a), with no sign error for the sum of the roots, and that a very large proportion of candidates were careful to insert a minus sign and 'equals zero' in

their equation in part (c). In part (b) (i) some candidates wrote down the correct expansion, while others worked it out laboriously from first principles. Some failed to understand the word 'Expand' and instead gave simply a numerical value. An even more unexpected mistake was to quote, or rather misquote, a formula for $\alpha^3 + \beta^3$ and write the absurd statement $(\alpha + \beta) = (\alpha + \beta) - 3\alpha\beta(\alpha + \beta)$. Luckily for these candidates, only one mark was lost and the rest of the question was often completed successfully.

Q16.

This question allowed most candidates to make a good confident start to the paper. Occasionally the values of $\alpha + \beta$ and $\alpha\beta$ were interchanged or one of them had the wrong sign. More seriously, in part (c), some candidates equated $(\alpha\beta)^3$ with $(\alpha + \beta)^3$ and often felt it necessary to expand $(\alpha + \beta)^3$, causing them a considerable loss of time. The mark scheme had only one mark for putting together the final equation. Hence, the distressingly common mistake of omitting the "= 0" at the end was not penalised on this occasion.

Q17.

There were many high-scoring attempts at this question, most candidates showing the necessary familiarity with the roots and coefficients of quadratic equations as well as with complex numbers. The explanation called for in part (a)(iii) often showed an awareness that real numbers cannot have negative squares, but failed to indicate the fact that $\alpha + \beta^2$ was not simply the square of α or of β . Some reference to the addition of the two squares was needed for full credit here.

In part (b) some candidates gave answers involving α and β rather than using the results they had obtained in part (a). In part (c) many candidates lost a mark by giving the wrong sign for the x term and/or omitting the 'equals zero' from their 'equation'.